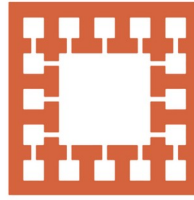


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Varendra University

Dept. of Computer Science & Engineering (CSE)

Microprocessor

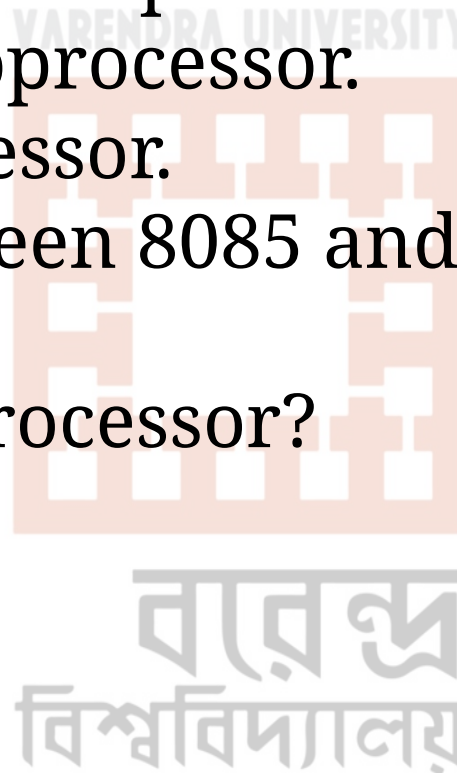
Course Title: **Microprocessor and Assembly Language**

Course No: **CSE 3205**

Prepared By:
Md. Adnan Sami
Lecturer, Dept. of CSE, VU

Outline

1. Generation of Microprocessor.
2. 8080/8085 Microprocessor.
3. 8086 Microprocessor.
4. Difference between 8085 and 8086 Microprocessor.
5. Pipelining.
6. What is 16-bit processor?



Generation of Microprocessor

First Generation (Between 1971 – 1973)

- PMOS technology.
- 4 bit processors (16 pins).
- 8 and 16 bit processors (40 pins).

Second Generation (During 1973)

- NMOS technology.
- Faster speed, Higher density.
- 4 / 8/ 16 bit processors (40 pins).
- **Intel 8085 (8 bit processor).**

Generation of Microprocessor

Third Generation (During 1978)

- HMOS technology.
- Faster speed, Higher packing density.
- 16 bit processors (40/ 48/ 64 pins)
- Easier to program.
- Dynamically relatable programs.
- Intel 8086 (16 bit processor).

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Generation of Microprocessor

Fourth Generation (During 1980)

- HCMOS technology.
- 32 bit processors.
- Physical memory space 2^{24} bytes = 16 Mb.
- Virtual memory space 2^{40} bytes = 1 Tb.
- Intel 80386.

Fifth Generation

- Pentium.

Generation of Microprocessor

Pentium (1993)

- 32 bit microprocessor.
- 64 bit data bus.
- 32 bit address bus.
- 4 GB main memory.
- 60,66,90 MHz.

Pentium Pro (1993)

- 32 bit microprocessor.
- 64 bit data bus.
- 36 bit address bus.
- 64 GB main memory.

Pentium II (1997)

- 32 bit microprocessor.
- 64 bit data bus.
- 36 bit address bus.
- 64 GB main memory.
- Starts at 266 MHz.

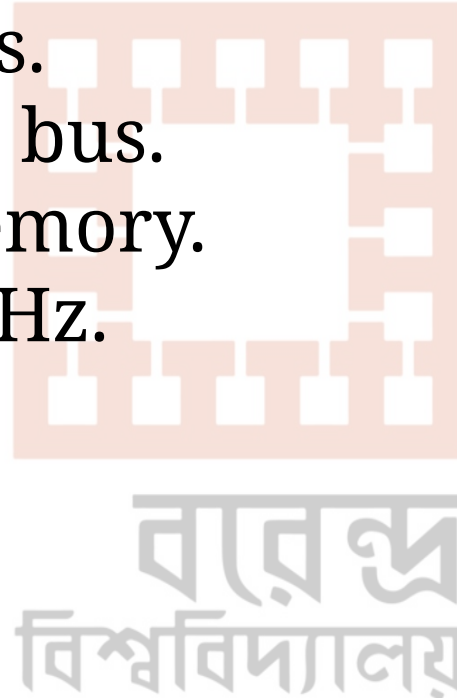
Pentium III (1999)

- 32 bit microprocessor.
- 64 bit data bus.
- 36 bit address bus.

Generation of Microprocessor

Pentium IV (2002)

- 32 bit microprocessor.
- 64 bit data bus.
- 32 bit address bus.
- 4 GB main memory.
- Starts at 1.4 GHz.



Generation of Microprocessor

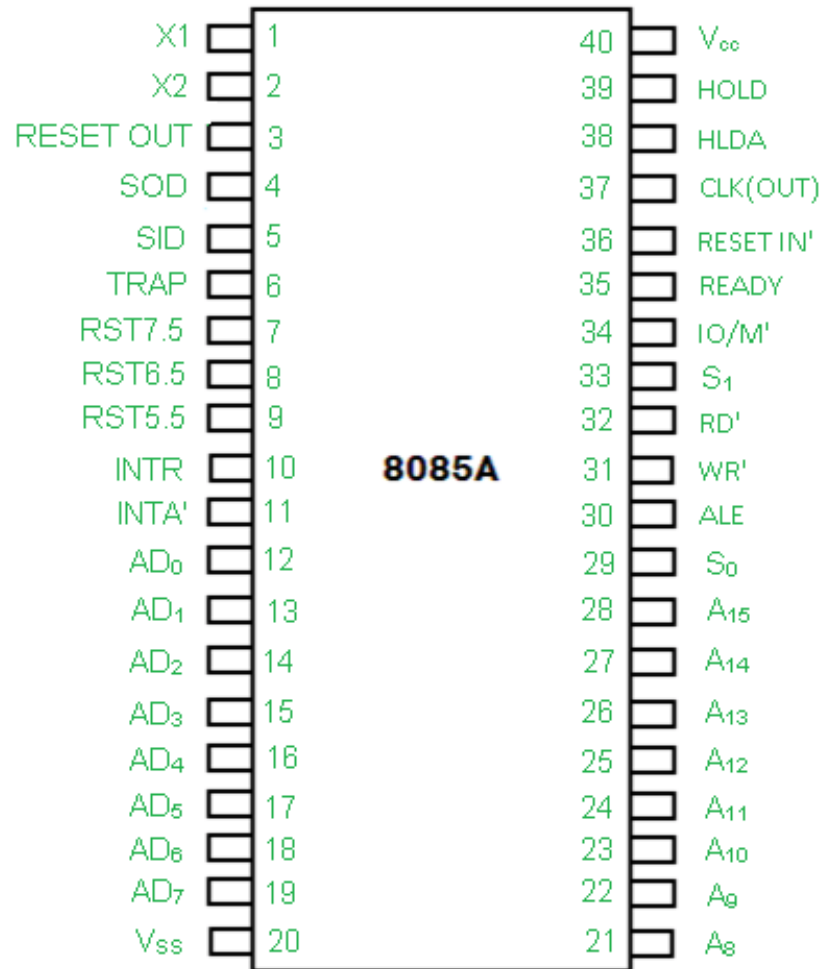
Processor	Year Intro.	Transistors	Clock Rate (MHz.)	External Data Bus	Internal Data Bus	Add. Bus
4004	1971	2,250	0.108	4	8	12
8008	1972	3,500	0.200	8	8	14
8080	1974	6,000	3	8	8	16
8085	1976	6,000	6	8	8	16
8086	1978	29,000	10	16	16	20
8088	1979	29,000	10	8	16	20
80286	1982	134,000	12.5	16	16	25
80386DX	1985	275,000	33	32	32	32
80386SX	1988	275,000	33	16	32	24
Pentium C	1993	3,100,000	66 -200	64	32	32
Pentium MMX	1997	4,500,000	300	64	32	32
Pentium Pro	1995	5,500,000	200	64	32	36
Pentium II	1997	7,500,000	233-450	64	32	36
Pentium III	1999	9,500,000	550-733	64	32	36
Itanium	2001	30,000,000	800-...	128	64	64

8080/8085 Microprocessor

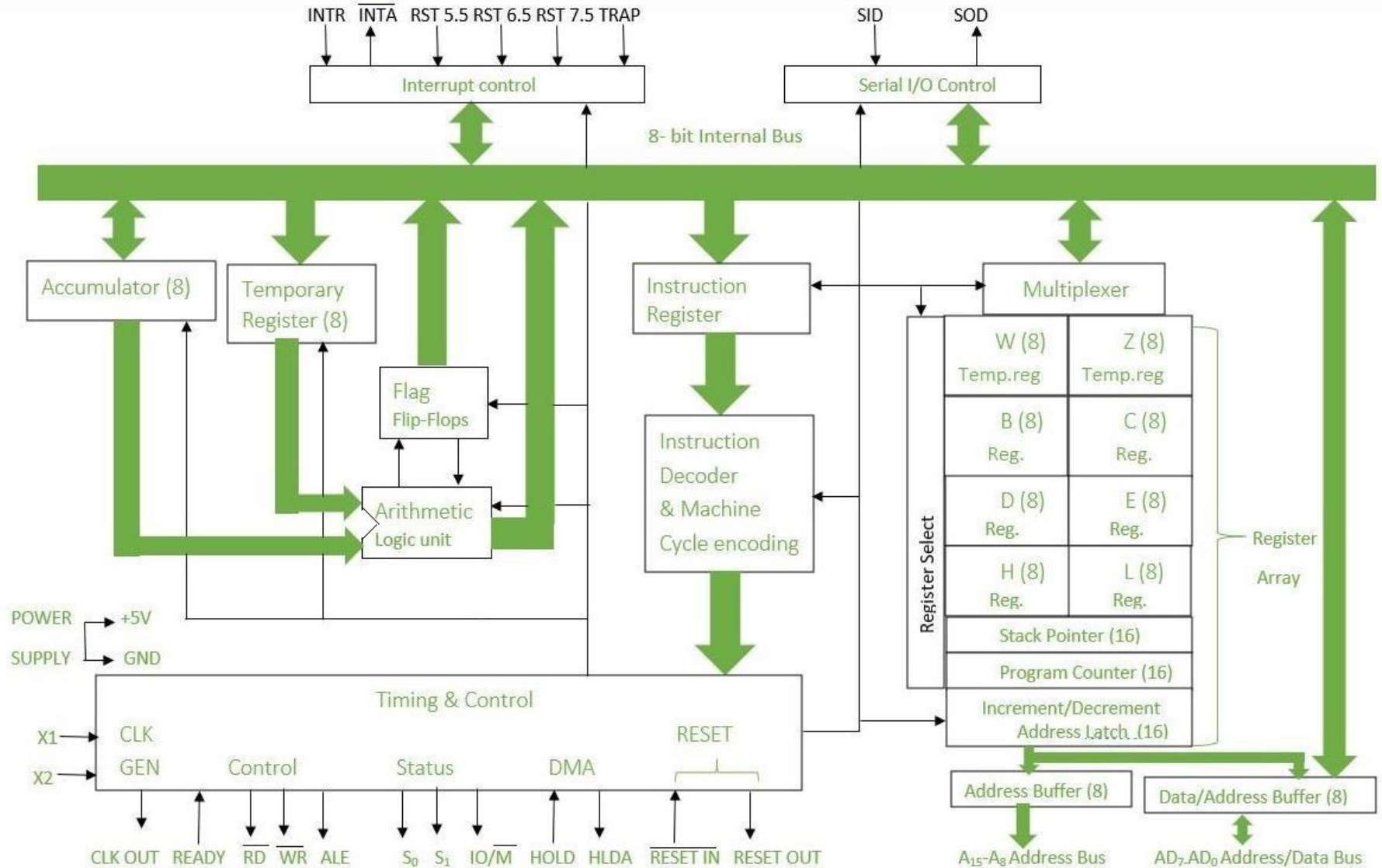
- 8 bit microprocessor.
- Developed by INTEL in the mid 1970.
- NMOS technology.
- Address bus 16 bit. $(2^{16} = 65,536 \text{ byte or } 64 \text{ KB})$
 $[65,536/1024 = 64]$
- Data bus 8 bit.
- **Non-pipelined processor.**
- It has 3 available clock speeds (3 MHz, 5 MHz, and 6 MHz).
- 8085 has memory capacity of 64 KB.
- It does not have an instruction queue.

8080/8085 Microprocessor

Pin diagram of 8085 microprocessor.



8080/8085 Microprocessor

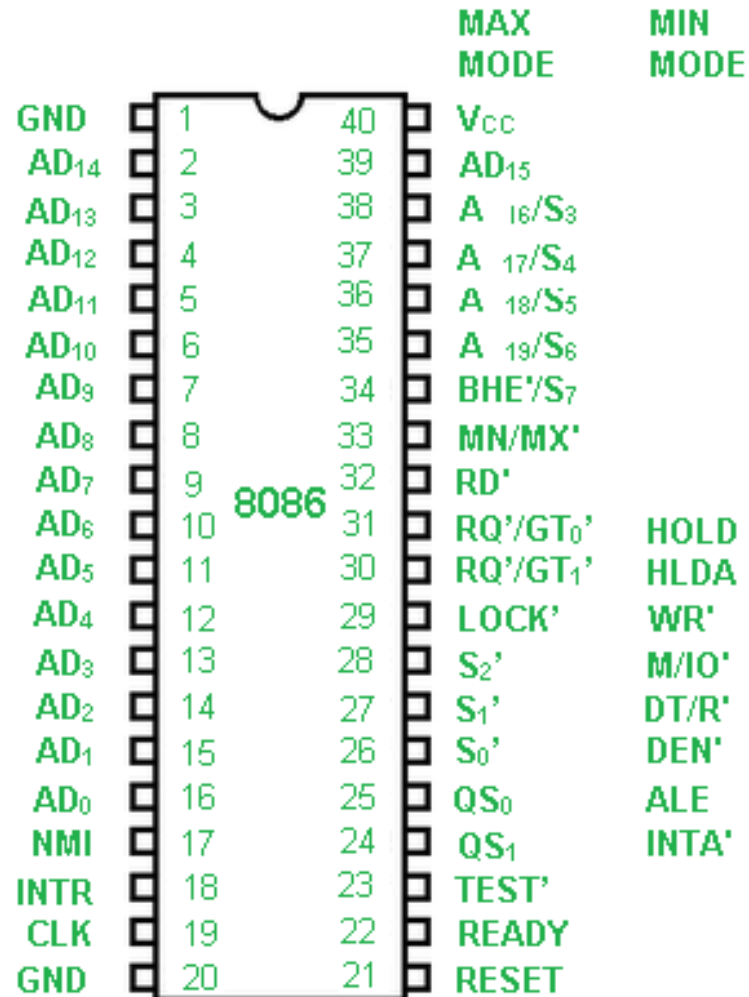


8086 Microprocessor

- 16 bit microprocessor.
- Developed by INTEL in the year 1978.
- Originally HMOS, now manufactured using HMOS III technique.
- Approximately 29, 000 transistors and 40 pin.
- 5V supply.
- 33% duty cycle.
- Address bus 20 bit. ($2^{20} = 1 \text{ MB}$)
- Data bus 16 bit.
- **Pipelined processor.**
- It has 3 available clock speeds (5 MHz, 8 MHz and 10 MHz).
- 8086 has memory capacity of 1 MB.
- It has 6 byte instruction queue.

8086 Microprocessor

Pin diagram of 8086 microprocessor.



Difference between 8085 and 8086

Microprocessor

8085	8086
8 bit microprocessor.	16 bit microprocessor.
Address bus 16 bit.	Address bus 20 bit.
Data bus 8 bit.	Data bus 16 bit.
Non-pipelined processor.	Pipelined processor.
Memory capacity 64 KB.	Memory capacity 1 MB.
NMOS technology.	HMOS III technology.

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Pipelining

The process of fetching the next instruction when the present instruction is being executed is called as pipelining.

Non-pipelined:

Fetch 1	Execute 1	Fetch 2	Execute 2	Fetch 3	Execute 3
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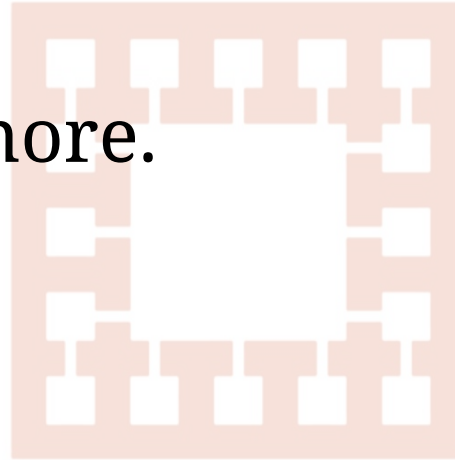
Pipelined:

Fetch 1	Execute		
	Fetch ¹ 2	Execute	
		Fetch ² 3	Execute
			3

Pipelining

Advantage of pipelining:

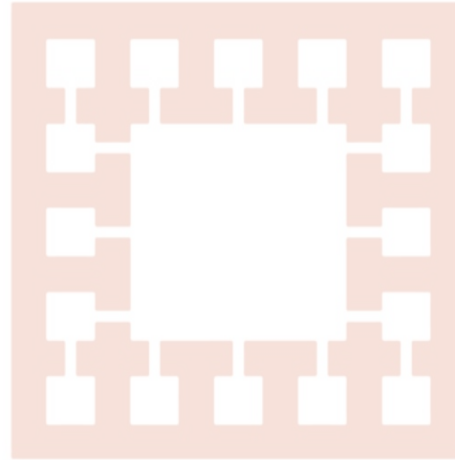
- Requires less time.
- Speed fast.
- Efficiency is more.



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What is 16-bit processor

A processor that can process 16-bit data at a time is called a 16-bit processor.



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